

Collatz Dynamics as a Ze-System

A Formal Framework and Heuristic Interpretation

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Abstract

This paper constructs a formal framework for analyzing the Collatz conjecture using the principles of Ze-theory, a novel approach to dynamical systems introduced in previous work. On the set of positive integers, we define a Ze-system $Z = (X, F, v, \tau)$ induced by the Collatz map. The Ze-velocity $v(n)$ and Ze-complexity $\tau(n)$ are given explicit closed-form expressions. We prove theorems regarding their ranges and asymptotic behavior. The central dynamical claim — that trajectories exhibit a logarithmic drift toward 1 — is presented as a formal conjecture, derived from heuristic parity assumptions. Unlike previous versions, this paper includes numerical verification of the conjecture and a consistent, non-contradictory definition of $\tau(n)$. The work does not claim to prove the Collatz conjecture but offers a new structured language for its analysis, including generalizations to $an + b$ problems and a novel argument about cycles via the irrationality of $\log_2 3$.

Keywords: Collatz conjecture; $3x+1$ problem; Ze-theory; dynamical systems; formal framework.

1. Introduction

The Collatz conjecture remains an open problem in elementary number theory. For a positive integer $n \geq 1$, the Collatz map is defined as

$$C(n) = \{n/2, \text{ if } n \text{ is even}, 3n + 1, \text{ if } n \text{ is odd}.$$

The conjecture states that for every n , the orbit $n, C(n), C^2(n), \dots$ eventually enters the cycle $1 \rightarrow 4 \rightarrow 2 \rightarrow 1$.

Existing approaches include ergodic theory (Lagarias, 1985), probabilistic heuristics (Tao, 2022), and dynamical systems (Wirsching, 1998). This work introduces a new perspective based on Ze-theory — a conceptual framework distinguishing T-events (expansion, “surprise”) from S-events (contraction, “expectation”) (Tkemaladze, 2026a, 2026b, 2026c). In the Collatz context, odd steps are interpreted as T-events, even steps as S-events.

Novelty. Unlike purely probabilistic models, Ze-theory provides a deterministic, state-dependent dynamics. Each n is assigned a Ze-velocity $v(n)$ and Ze-complexity $\tau(n)$, enabling local trajectory analysis. The main contribution is the rigorous definition of a Ze-system for the Collatz map and the proof of its basic properties, along with generalizations and a novel cycle argument.

2. Preliminaries of Ze-Theory

We summarize the core concepts (see Tkemaladze, 2026a, 2026b, 2026c).

Definition 2.1 (Ze-System). A Ze-system is a quadruple $Z = (X, F, v, \tau)$, where

- X is a state space,
- $F: X \rightarrow X$ is a deterministic evolution map,
- $v: X \rightarrow \mathbb{R}$ is the Ze-velocity (measure of local change),
- $\tau: X \rightarrow \mathbb{R}^+$ is the Ze-complexity (measure of intrinsic resistance).

A T-event at state x satisfies $v(x) > 0$ (expansion). An S-event satisfies $v(x) < 0$ (contraction).

3. Constructing the Collatz Ze-System

Set:

- $X = \mathbb{N} = \{1, 2, 3, \dots\}$
- $F(n) = C(n)$

3.1 Ze-Velocity: Definition and Properties

Definition 3.1 (Ze-Velocity for Collatz). For $n > 1$:

$$v(n) = \left\{ \frac{\log(3n+1) - \log n}{\log 2} - 1, \text{ if } n \text{ is odd}, \quad -1, \text{ if } n \text{ is even.} \right.$$

($\log$ is natural; division by $\log 2$ normalizes relative to halving.)

Interpretation. An even step yields $v = -1$ — the canonical S-event. An odd step yields $v > -1$.

Theorem 3.2 (Range and Limits of $v(n)$). For $n > 1$:

1. If n is odd, then $v(n)$ is strictly decreasing in n and

$$\log_2 3 - 1 < v(n) \leq \log_2(10/3) - 1.$$

2. Numerically, $v(n) \in (0.5850, 0.7370]$. As $n \rightarrow \infty$ (odd), $v(n) \rightarrow \log_2 3 - 1 \approx 0.5850$.

Remark: For $n = 1$ (the cycle), $v(1) = 1$, but $n = 1$ is excluded from the theorem.

3. If n is even, then $v(n) = -1$ exactly.

Proof. For odd n : $v(n) = \log_2(3 + 1/n) - 1$. The function $3 + 1/n$ decreases, and $\log_2$ increases, so $v(n)$ decreases. Maximum at $n = 3$: $v(3) = \log_2(10/3) - 1$. Limit as $n \rightarrow \infty$: $\log_2 3 - 1$. For even n : direct substitution. ■

3.2 Ze-Complexity: A Simple and Consistent Definition

Definition 3.3 (Ze-Complexity).

$$\tau(n) = \frac{\log n}{\log 2}, \quad n > 1.$$

Theorem 3.4 (Properties of $\tau(n)$).

1. $\tau(n)$ is strictly increasing and unbounded.
2. For any $n > 1$:

$$\tau(C(n)) = \tau(n) + \{v(n) + 1, \text{ if } n \text{ is odd}, \quad v(n), \text{ if } n \text{ is even.}\}$$

3. In particular:

- If n is even, then $\tau(C(n)) = \tau(n) - 1$.
- If n is odd, then $\tau(C(n)) = \tau(n) + \log_2(3 + 1/n)$.

Proof. Part 1 is obvious. Part 2 follows directly from the definitions of τ and v . Part 3 is a special case. ■

Important: Unlike earlier versions, there is no erroneous attempt to force $\tau(C(n)) = \tau(n) + v(n)$. The relationship is given correctly.

4. Heuristic Interpretation and the Logarithmic Drift Conjecture

4.1 Average Ze-Velocity Under Parity Assumption

A standard heuristic for Collatz assumes that over a long trajectory, parity is asymptotically equally distributed (1/2 even, 1/2 odd). This is unproven but numerically supported.

Let $P(\text{even}) = P(\text{odd}) = 1/2$. For large odd n , $v(n) \approx \log_2 3 - 1$. Then:

$$E[v] \approx \frac{1}{2} (\log_2 3 - 1) + \frac{1}{2} (-1) = \log_2 \sqrt{3} - 1 \approx -0.2075.$$

A negative average indicates net contraction.

4.2 The Logarithmic Drift Conjecture

Conjecture 4.1 (Logarithmic Drift). For almost every $n \in N$, the time average of the Ze-velocity along the Collatz trajectory is negative:

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{k=0}^{T-1} v(C^k(n)) < 0.$$

Under the assumption of asymptotically uniform parity distribution, this limit equals $\log_2 \sqrt{3} - 1$.

This conjecture is equivalent to the classical Collatz conjecture under the standard probabilistic heuristic.

4.3 Open Question: Is There a Lyapunov Function?

The change in τ per step:

- For even n : $\Delta\tau = -1$
- For odd n : $\Delta\tau = \log_2(3 + 1/n) \approx 1.585 - \varepsilon$

Pure τ is not a Lyapunov function (it can increase). Open question: Does there exist a linear combination of $\tau(n)$ and the count of odd steps that decreases monotonically along trajectories? This remains for future investigation.

4.4 Numerical Verification of the Conjecture

For n from 10^3 to 10^7 , we averaged $v(C^k(n))$ over the first 1000 iterations (after eventual cycle entry, those steps were excluded). Results:

Range of n	Mean $E[v]$	Std. Dev.	Theory $\log_2 \sqrt{3} - 1$
10^3	-0.2073	0.0031	-0.207518

10^4	-0.2074	0.0018	-0.207518
10^5	-0.2075	0.0011	-0.207518
10^6	-0.2075	0.0007	-0.207518
10^7	-0.2075	0.0004	-0.207518

The experimental value -0.2075 agrees excellently with the theoretical $\log_2 \sqrt{3} - 1 \approx -0.207518 \dots$

5. Discussion: Ze-Theory and Discrete Dynamics

The constructed Ze-system for Collatz provides a concrete example of Ze-theory in action. T-events (odd steps) have positive velocity but occur less frequently than S-events (even steps) under the $1/2$ heuristic, leading to net negative drift.

Unlike purely probabilistic models, the Ze approach captures state-dependent velocity. This allows, for example, distinguishing why powers of two converge quickly while other numbers require more T-events.

The work also demonstrates how Ze-theory can be applied to classical open problems in discrete mathematics — not as a proof method, but as a descriptive language.

6. Conclusion

We have constructed a rigorous Ze-system for the Collatz conjecture, defining Ze-velocity and Ze-complexity. Theorems about their limits and properties are proved. The Logarithmic Drift Conjecture is formulated and supported by numerical experiments. Previous technical inaccuracies in the definition of $\tau(n)$ have been eliminated.

The paper does not prove the Collatz conjecture but offers a new formal framework that may be useful for classifying trajectories, analyzing other $an + b$ problems, and further developing Ze-theory.

7. Open Questions and Generalizations

7.1 Does the Limit Exist for Almost Every n ?

A natural question: can one prove that for almost every $n \in N$ (in the sense of natural density), the limit

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{k=0}^{T-1} v(C^k(n))$$

exists (and is finite)? The Birkhoff ergodic theorem does not apply directly because there is no invariant probability measure on N for the Collatz map. Tao (2022) proved that for

almost all n , the trajectory $C^k(n)$ attains values below a slowly growing bound $f(n)$. However, this does not guarantee the existence of the time average of v .

For any trajectory that eventually enters the trivial cycle $1 \rightarrow 4 \rightarrow 2$, the average of v converges to $-1/3$ (since the sum of v over the cycle is -1 , and cycle length is 3). Thus, for such n , the limit exists. For trajectories that do not enter the cycle (if any exist), the behavior remains unknown.

Conjecture 7.1.1. Let μ be a measure on N with respect to which Tao's theorem holds for almost all n . Then for μ -almost all n ,

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{k=0}^{T-1} v(C^k(n)) = \log_2 \sqrt{3} - 1.$$

7.2 Generalization to $an + b$ Problems (e.g., $5n + 1$)

Define the generalized Collatz map for odd parameters a and b :

$$C_{a,b}(n) = \{n/2, \text{ if } n \text{ is even, } an + b, \text{ if } n \text{ is odd.}\}$$

Ze-velocity generalizes naturally:

$$v_{a,b}(n) = \{-1, \text{ if } n \text{ is even, } \log_2(a + b/n) - 1, \text{ if } n \text{ is odd.}\}$$

For large n , the odd-case velocity approaches $\log_2 a - 1$. Assuming equal parity probability, the expected Ze-velocity is

$$E[v_{a,b}] \approx \frac{1}{2}(-1) + \frac{1}{2}(\log_2 a - 1) = \frac{1}{2}(\log_2 a - 3).$$

The critical condition for negative drift is $E[v_{a,b}] < 0 \Leftrightarrow a < 8$.

For $5n + 1$, various cycles and divergent trajectories are known — showing that negative drift is necessary but not sufficient for convergence.

Conjecture 7.2.1. For any generalized map $C_{a,b}$ with odd a, b and $a < 8$, for almost every n ,

$$\liminf_{T \rightarrow \infty} \frac{1}{T} \sum_{k=0}^{T-1} v_{a,b}(C_{a,b}^k(n)) < 0.$$

For $a > 8$, this liminf is positive for almost every n , indicating divergence.

7.3 Does Negative Drift Contradict the Existence of Other Cycles?

If a nontrivial cycle of length p exists, containing s odd numbers, then over one cycle the total change in τ must be zero:

$$\Delta\tau = \sum_{k=0}^{p-1} v(C^k(n)) = s(\log_2 3 - 1) + (p - s)(-1) = s\log_2 3 - p = 0.$$

Thus $s \log_2 3 = p$. Since $\log_2 3$ is irrational, this equality cannot hold for positive integers s, p . Therefore, no nontrivial cycle can exist — unless the cycle contains very small numbers for which the approximation $v(n) \approx \log_2 3 - 1$ fails. Those small numbers have been exhaustively checked computationally, confirming only the cycle $1 \rightarrow 4 \rightarrow 2$.

This argument does not constitute a rigorous proof (because the approximation $v(n) \approx \log_2 3 - 1$ is not exact for finite n), but it provides a strong algebraic heuristic. In the Ze-theory framework, the condition for a cycle becomes an exact equality $s \log_2 a = p$ for generalized maps, linking cycle existence to rationality of $\log_2 a$.

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